```
0010  REM THIS PROGRAM CALCULATES THE SUGAR AND MOLASSES IN STOCK IN GALLONS CALLED "MRSTOCKG"
0020  DIM A$(18)32
0030  FOR I=1 TO 18
0040  FOR J=1 TO 12
0050  E(I,J)=0
0060  NEXT J
0070  NEXT I
0080  A$(1)="Juice"
0090  A$(2)="Syrup"
0100  A$(3)="A Massecuite"
0105  A$(4)="Top-Off A Mass."
0110  A$(5)="B Massecuite"
0120  A$(6)="C Massecuite"
0130  A$(7)="A Molasses"
0135  A$(8)="Top-Off A Mol."
0140  A$(9)="B Molasses"
0150  A$(10)="Grain"
0155  A$(11)="B Magma"
0160  A$(12)="C Magma"
0170  A$(13)="Other"
0180  A$(14)="TOTAL"
0190  A$(15)="Final Molasses Purity"
0200  A$(16)="Sugar Purity"
0210  A$(17)="Sugar in Stock"
0220  A$(18)="Final Molasses in Stock"
0230  INPUT "1   JUICE, GALLONS",E(1,1)
0240  INPUT "2   JUICE, BRIX",E(1,2)
0250  INPUT "3   JUICE, POL",E(1,3)
0260  INPUT "4   SYRUP, GALLONS",E(2,1)
0270  INPUT "5   SYRUP, BRIX",E(2,2)
0280  INPUT "6   SYRUP, POL",E(2,3)
0290  INPUT "7   A MASSECUITE, GALLONS",E(3,1)
0300  INPUT "8   A MASSECUITE, BRIX",E(3,2)
0310  INPUT "9   A MASSECUITE, POL",E(3,3)
0312  INPUT "10  TOP-OFF A MASSECUITE, GALLONS",E(4,1)
0314  INPUT "11  TOP-OFF A MASSECUITE, BRIX",E(4,2)
0316  INPUT "12  TOP-OFF A MASSECUITE, POL",E(4,3)
0320  INPUT "13  B MASSECUITE, GALLONS",E(5,1)
0330  INPUT "14  B MASSECUITE, BRIX",E(5,2)
0340  INPUT "15  B MASSECUITE, POL",E(5,3)
0350  INPUT "16  C MASSECUITE, GALLONS",E(6,1)
0360  INPUT "17  C MASSECUITE, BRIX",E(6,2)
0370  INPUT "18  C MASSECUITE, POL",E(6,3)
0380  INPUT "19  A MOLASSES, GALLONS",E(7,1)
0390  INPUT "20  A MOLASSES, BRIX.",E(7,2)
0400  INPUT "21  A MOLASSES, POL.",E(7,3)
0402  INPUT "22  TOP-OFF A MOLASSES, GALLONS",E(8,1)
0404  INPUT "23  TOP-OFF A MOLASSES, BRIX",E(8,2)
0406  INPUT "24  TOP-OFF A MOLASSES, POL",E(8,3)
0410  INPUT "25  B MOLASSES, GALLONS",E(9,1)
0420  INPUT "26  B MOLASSES, BRIX",E(9,2)
0430  INPUT "27  B MOLASSES, POL",E(9,3)
0440  INPUT "28  GRAIN, GALLONS",E(10,1)
0450  INPUT "29  GRAIN, BRIX",E(10,2)
0460  INPUT "30  GRAIN, POL",E(10,3)
0462  REM INPUT "31  B MAGMA, GALLONS",E(11,1)
0464  REM INPUT "32  B MAGMA, BRIX",E(11,2)
0466  REM INPUT "33  B MAGMA, POL",E(11,3)
0470  INPUT "34  C MAGMA, GALLONS",E(12,1)
0480  INPUT "35  C MAGMA, BRIX",E(12,2)
0490  INPUT "36  C MAGMA, POL",E(12,3)
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0500  INPUT "37  OTHER MATERIAL, GALLONS",E(13,1)
0510  INPUT "38  OTHER MATERIAL, BRIX",E(13,2)
0520  INPUT "39  OTHER MATERIAL, POL",E(13,3)
0530  INPUT "40  FINAL MOLASSES PURITY FOR RUN",E(15,1)
0540  INPUT "41  SUGAR PURITY FOR RUN",E(16,1)
0550  FOR J=1 TO 13
0560  IF E(J,2)=0 THEN 620
0570  E(J,4)=E(J,3)*100/E(J,2)
0580  E(J,5)=(62.2511+.24081*E(J,2)+.0007902404*E(J,2)^2+.00000423954*E(J,2)^3-1.657193E-8*E(J,2)^4)/7.48
0590  E(J,6)=E(J,1)*E(J,5)/2000
0600  E(J,7)=E(J,6)*E(J,2)/100
0610  E(J,8)=E(J,6)*E(J,3)/100
0620  NEXT J
0630  FOR J=1 TO 8
0640  E(14,J)=E(1,J)+E(2,J)+E(3,J)+E(4,J)+E(5,J)+E(6,J)+E(7,J)+E(8,J)+E(9,J)+E(10,J)+E(11,J)+E(12,J)+E(13,J)
0650  NEXT J
0660  E(14,2)=E(14,7)*100/E(14,6)
0670  E(14,3)=E(14,8)*100/E(14,6)
0680  E(14,4)=E(14,3)*100/E(14,2)
0690  E(14,5)=E(14,6)*2000/E(14,1)
0700  E(17,1)=E(14,7)*(E(14,4)-E(15,1))/(E(16,1)-E(15,1))
0710  E(17,2)=E(17,1)*E(16,1)/100
0720  E(18,1)=E(14,7)-E(17,1)
0730  E(18,2)=E(14,8)-E(17,2)
0740  E(17,3)=E(17,2)*2000/.96
0750  K1=(62.2511+.24081*H(30)+.0007902404*H(30)^2+.00000423954*H(30)^3-1.657193E-8*H(30)^4)/7.48
0760  K1=K1*H(30)/100
0770  E(18,3)=E(18,1)*2000/K1
0780  SELECT PRINT 215(132)
0782 IF H(36)=0 THEN 790
0784 SELECT PRINT 204(132)
0790  IF H(23)=0 THEN 820
0800  PRINT HEX(0E);TAB(33-LEN(B$)/2);B$
0810  GOTO 830
0820  PRINT TAB(66-LEN(B$)/2);B$
0830  PRINT HEX(0A0A)
0840 IF H(23)=0 THEN 870
0850  PRINT HEX(0E);TAB(24);"STOCK CALCULATIONS"
0860 GOTO 880
0870 PRINT TAB(57);"STOCK CALCULATIONS"
0880  PRINT HEX(0A0A)
0890  PRINT "Date: ";A1$
0900  PRINT "--------------"
0910  PRINT HEX(0A0A)
0920  PRINT "Material in Stock     Gallons Measured     Brix     Pol     Purity    Pounds per Gallon     Tons Material     Tons Brix
    Tons Pol"
0930  PRINT "-----------------     ----------------    -----    -----    ------    -----------------     -------------     ---------
    --------"
0940  FOR J=1 TO 13
0950  PRINTUSING 960,A$(J),E(J,1),E(J,2),E(J,3),E(J,4),E(J,5),E(J,6),E(J,7),E(J,8)
0960  %################    -##,###,###.##     -###.##  -###.##   -###.##      -#,###.###           -#,###.###     -#,###.###   -#,#
##.###
0970  NEXT J
0980  PRINT "-------------           ----------       -----    -----    ------          ------              --------        -------
     -------"
0990  J=14
1000  PRINTUSING 960,A$(J),E(J,1),E(J,2),E(J,3),E(J,4),E(J,5),E(J,6),E(J,7),E(J,8)
1010  PRINT "-------------           ----------       -----    -----    ------          ------              --------        -------
     -------"
1020  PRINT HEX(0A0A0A0A0A)
1030  PRINT "Assumptions"
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1040  PRINT "-----------"
1050  PRINT HEX(0A)
1060  FOR J=15 TO 16
1070  PRINTUSING 1080,A$(J),E(J,1)
1080  %################################      -###.##
1090  NEXT J
1100  PRINT HEX(0A0A0A0A0A)
1110  PRINT "                                                  SUGAR AND MOLASSES IN PROCESS"
1120  PRINT "                                                  ----------------------------"
1130  PRINT HEX(0A0A)
1140  PRINT "                                    Lbs Brix           Lbs Pol"
1150  PRINT "                                    --------           -------"
1160  PRINTUSING 1170,A$(17),E(17,1)*2000,E(17,2)*2000,E(17,3)
1170  %########################       -###,###,###      -###,###,###           (Lbs. 96 Sugar in Stock = -###,###.##)
1180  PRINT HEX(0A)
1190  PRINTUSING 1200,A$(18),E(18,1)*2000,E(18,2)*2000,H(30),E(18,3)
1200  %########################       -###,###,###      -###,###,###           (Final Molasses in Stock - Gals. @ ##.## Brix =-###,#
##.##)
1210  PRINT "                                   ---------          --------"
1220  PRINTUSING 1230,E(14,7)*2000,E(14,8)*2000
1230  %TOTAL                         -###,###,###      -###,###,###
1240  PRINT "                                   ---------          --------"
1250  PRINT HEX(0C)
1260  SELECT PRINT 005(64)
1270  INPUT "IS DATA OKAY (Y OR N)",J$
1280  IF J$="Y" THEN 1310
1290  IF J$="N" THEN 230
1300  GOTO 1270
1310  LOAD T "MRSTART" 10
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